5.0

TECHNICAL DOCUMENTATION SERIES

DIGITAL TWIN & PROCESS CATALOGUE

PROJECT:
SARL ALLO MAISON

DATE:
DEC 2025

01 PROCESS & WORKFLOW

OPERATIONAL OVERVIEW

Le projet ALLO MAISON vise à créer un complexe industriel pour la production de matériaux polymères et composites, notamment des panneaux polymères, des revêtements de sol, des portes intérieures et extérieures, des panneaux muraux décoratifs, des pièces thermoformées et des articles moulés par injection et extrusion plastique. Le projet s'inscrit dans le cadre de l'industrie 5.0, avec une automatisation intégrale et une utilisation de technologies avancées telles que la robotique et l'intellige...



Fig 1.1: General Layout

KEY OBJECTIVES:

- Substitution Aux Importations
- Création De Valeur Locale
- Export Potentiel
- Transfert Technologique

CORE PILLARS

1. AUTOMATION
Automated systems ensure consistent quality and reduced cycle times. **2. CONNECTIVITY**
Integrated IoT sensors provide real-time monitoring and data analytics. **3. SUSTAINABILITY**
Energy-efficient processes minimize environmental impact.

02 PROCESS ARCHITECTURE

WORKFLOW DESCRIPTION

The production workflow follows a sequential process flow optimized for throughput and quality control.



Each stage is monitored for quality assurance and process optimization.

03 MACHINE ASSETS

This facility operates 10 primary production assets.

MACHINE REF-01 CATEGORY A



Manufacturer	Brand Name
Capacity	000 Units/Hr
Power	00 kW
Status	**Operational**

Sed commodo posuere pede. Mauris ut est.

MACHINE REF-02 CATEGORY B



Manufacturer	Brand Name
Capacity	000 Units/Hr
Power	00 kW
Status	**Maintenance**

Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Donec odio elit, dictum in, hendrerit sit amet, egestas sed, leo.

04 PRODUCT CATALOG

Primary product: PET Bottled Water (0.5L) with target output of 12000 bottles/hour.

REF ID	PRODUCT NAME	MATERIAL COMPOSITION	CATEGORY
A-001	**Product Name Alpha**	Lorem Ipsum Material	Type A
B-002	**Product Name Beta**	Dolor Sit Amet	Type B
C-003	**Product Name Gamma**	Consectetur Adipiscing	Type C
D-004	**Product Name Delta**	Elit Sed Do	Type D

05 LOGISTICS & SUPPLY CHAIN

INBOUND FLOW

Raw materials are sourced from certified suppliers following quality protocols.

- Metric A: 0000 Units
- Metric B: 00% Efficiency

OUTBOUND FLOW

Finished products are dispatched through optimized logistics channels.

- Metric C: 0000 Pallets
- Metric D: 00h Lead Time



06 DIGITAL TWIN INTERFACE

INTERFACE GUIDE: TWIN MANAGER v5.0

The Digital Twin interface provides a real-time, browser-based supervision of the factory floor. It is built using **React.js** and **Three.js** to render a 3D interactive scene alongside critical live data.

1. 3D VISUALIZATION (CANVAS)

The Dashboard features a `ComplexFactoryScene` that renders the physical layout:

- **Assets:** Displays CNC Machines, Robot Arms, and Conveyor Belts.
- **Status Indicators:** Machines change color dynamically based on their health state: **Green (Running)**, **Amber (Idle)**, or **Red (Warning)**.
- **Animation:** Robot arms and AGVs (Automated Guided Vehicles) move in sync with the production cycle time.

2. PREDICTIVE MAINTENANCE (SCATTER CHART)

The Maintenance module utilizes a **Risk Matrix Scatter Chart** to correlate three key variables:

- **X-Axis:** Temperature (°C).
- **Y-Axis:** Machine Health (%).
- **Z-Axis (Point Size):** Vibration levels.

Logic: The system triggers a `status="warning"` if Temperature > 90°C. This allows operators to visualize outliers (machines that are hot and vibrating) before failure occurs.

3. TCO OPTIMIZATION (BAR CHART)

The **Total Cost of Ownership (TCO)** calculator projects financial data over 10 years based on live telemetry.

- **Calculation:** `Total = CAPEX + (Power * Hours * Cost) + Maintenance`.
- **Stacked Bar Chart:** Visualizes the accumulation of OPEX (Energy + Maintenance) on top of the initial CAPEX investment.
- **User Inputs:** Financial directors can adjust Energy Cost (€/kWh) and Years to simulate different ROI scenarios.

4. SAFETY & ATEX MONITORING

For hazardous zones (e.g., Painting Robot), the system monitors air quality.

- **Data Viz:** An Area Chart tracks PPM (Parts Per Million) of VOCs/Solvents in real-time.
- **Automation:** If concentration spikes, ventilation status switches to "100%" instantly.